

External Evidence and Language-Model Forecasts: A Blinded Comparison with Prediction-Market Prices

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Abstract

We use prediction-market prices as prospective intermediate benchmarks for evaluating language-model forecasts. Unresolved future events avoid leakage of realized outcomes, while market prices provide immediate feedback from an external, financially incentivized forecasting mechanism. Using the Vorhersage workbench, we compare question-only and evidence-supplied forecasts for six unresolved Kalshi contracts across six subject areas. The model, instructions, and response schema are held fixed, with one independent call per contract and condition. Opening quotes are withheld until forecasts are sealed, and Kalshi is excluded as a source of research evidence. Mean absolute deviation from the opening bid-ask midpoint is 33.17 percentage points in the question-only condition and 8.58 points in the research condition. Research is associated with a smaller deviation in five cases and an unchanged deviation in one. These results demonstrate improved agreement with an economically meaningful intermediate benchmark. The small purposive sample and single draw per condition limit generalization; a documented entity-identification error also persists in the Netflix forecast. The design supports rapid, reproducible comparisons before settlement, which can subsequently be extended with outcome scoring.

1 Introduction

A forecasting system must translate evidence about an event into a numerical probability. Liquid prediction markets provide a useful intermediate benchmark for evaluating those probabilities before events resolve. This study asks whether supplying current, relevant facts moves a model's forecasts closer to that benchmark by comparing two information conditions on the same set of unresolved prediction-market contracts. One condition supplies only the event definition; the other adds a frozen packet of primary-source evidence. Both use the same language model and forecasting instructions.

There is a substantial empirical basis for this choice. Wolfers and Zitzewitz [1] document accurate market forecasts across several settings and favorable comparisons with conventional benchmarks. Page and Clemen [2] find reasonably well-calibrated probabilities for shorter-horizon contracts, alongside horizon-dependent biases. In a distinct test of forecasting accuracy, Berg, Nelson, and Rietz [3] find that Iowa Electronic Markets vote-share forecasts are closer to eventual election results than 74% of 964 contemporaneous polls. The latter is evidence about vote-share accuracy, rather than calibration of binary event probabilities. Wolfers and Zitzewitz [4] also provide theoretical conditions under which prices recover average beliefs, and show useful approximations under broader assumptions. Together, this literature motivates well-traded, clearly specified contracts as informative probability benchmarks, with attention to liquidity and forecast horizon.

The prospective design has an additional advantage for evaluating language models. For a genuinely unresolved future event, the eventual realized outcome does not yet exist to be memorized in training data or retrieved from a later account. Evaluating at that point avoids the realized-outcome leakage that can affect historical forecasting exercises, without requiring reconstruction of a historical information set. Market quotes make evaluation possible immediately, rather than only after all events settle. Exposure to the market price is a separate issue: this experiment withholds target quotes and excludes market commentary from the forecasting inputs. The measured quantity is *market agreement*, interpreted as performance against an empirically motivated intermediate probability benchmark.

Vorhersage provides the registration, evidence storage, forecast validation, and separation of hidden targets from model inputs. The present experiment uses a minimal direct-judgment method to isolate the information contrast as far as possible. It does not compare the package’s full structured workflow with an unstructured alternative. The dated protocol, registration, and result artifacts provide the empirical record for the analysis [5, 6].

2 Experimental design

2.1 Contract selection and target measurement

The sample consists of six contracts selected to span weather, employment, monetary policy, space technology, entertainment, and sports. Candidate contracts and ordered substitutes were recorded before order-book retrieval. Previously revealed cases were excluded. Eligibility required an active, unresolved binary contract; an uncrossed, non-extreme two-sided book; at least one contract at each best quote; and a bid–ask spread no greater than ten cents. Two initial candidates failed these requirements and were replaced by predefined substitutes. The researcher received eligibility flags without seeing accepted or rejected quotes. This procedure limits direct selection on observed prices, although quote eligibility still conditions the sample. The cohort was selected purposively rather than sampled randomly.

Opening books were captured on September 18, 2026, between 11:06:23 and 11:06:26 UTC (7:06 a.m. EDT), before external research. The reference value is the arithmetic midpoint of the best YES bid and ask. Equal weight is assigned to each contract, irrespective of quoted size, topic, or forecast horizon. Table 1 identifies the events and recorded probabilities.

2.2 Information conditions and forecasting procedure

An experimental *arm* in Vorhersage binds a versioned method, model configuration, and data assignment. Both arms used Google’s *gemini-3.1-pro-preview* with identical instructions and output schemas. The question-only arm received the precise event and settlement conditions. The research arm received those inputs plus a frozen packet containing source excerpts, dates, extracted facts, and limitations. An external coding agent selected and collected the evidence from official sources; the forecasting model itself had no browsing tools or local filesystem access. Across the six packets, nine source records were supplied. Appendix B describes their contents and omissions.

Each contract received one fresh model call per arm, producing twelve forecasts. Both arms were generated after evidence collection, in separate contexts. Consequently, the question-only estimate is an information ablation, not a chronological pre-research forecast. The common task requested a probability, a concise rationale, reasons the estimate could be too high or too low, and

limitations. The method had an assessment stage followed by deterministic issuance, with no model review or adjudication. Full settings and execution details appear in Appendix A.

2.3 Blinding and provenance

Kalshi supplied contract definitions and evaluator targets, but no Kalshi material was used as research evidence. The package exposed only allowlisted contract-definition fields to the researcher. Opening order books were saved in a separate evaluator database; the research project stored commitments to the hidden targets. Research was restricted to approved primary-source domains. URL and redirect checks, together with lexical screening of returned text, were applied before display to exclude prediction-market and betting-odds material. Forecasting prompts contained only the assigned question, common instructions, and permitted evidence.

All twelve forecasts were sealed at 11:25:10 UTC. Targets were revealed between 11:40:25 and 11:40:27 UTC, after the study’s reveal routine verified an unchanged forecast seal and a terminal attempt for every registered trial. The subsequent audit checked target commitments, evidence and forecast hashes, and recorded timing [7]. The design provides auditable procedural blinding: the exact hosted-model inputs are inspectable, although the coordinator was not isolated from the evaluator database by operating system permissions. Remaining leakage concerns involve market prices or market-derived commentary, which lexical filters may not always detect. These are distinct from leakage of the future realized outcome, which is unavailable at the time of this prospective exercise.

2.4 Evaluation measures

Let $i = 1, \dots, n$ index contracts, with $n = 6$, and let $k \in \{Q, R\}$ denote the question-only and research conditions. Write b_i and a_i for the opening YES bid and ask, $q_i = (b_i + a_i)/2$ for their midpoint, and p_{ik} for the model forecast. The primary descriptive measure is mean absolute deviation from the midpoint, expressed in percentage points:

$$\text{MAE}_k = \frac{100}{n} \sum_{i=1}^n |p_{ik} - q_i|. \quad (1)$$

We also report root mean squared deviation and mean distance outside the opening bid–ask interval:

$$\text{RMSE}_k = 100 \sqrt{\frac{1}{n} \sum_{i=1}^n (p_{ik} - q_i)^2}, \quad (2)$$

$$\text{Outside}_k = \frac{100}{n} \sum_{i=1}^n \max\{b_i - p_{ik}, p_{ik} - a_i, 0\}. \quad (3)$$

The paired improvement for contract i is $\Delta_i = 100(|p_{iQ} - q_i| - |p_{iR} - q_i|)$; positive values indicate closer agreement in the research arm. A constant forecast of 0.5 provides an additional reference. The bid–ask interval describes quoted trading prices, not a statistical confidence interval.

Squared distance from a probability benchmark has a direct scoring interpretation. In the ideal case where $q_i = \mathbb{E}[Y_i | \mathcal{I}_i]$ is the event probability conditional on the relevant information set and p_{ik} is measurable with respect to that set,

$$\mathbb{E}[(p_{ik} - Y_i)^2 - (q_i - Y_i)^2 | \mathcal{I}_i] = (p_{ik} - q_i)^2. \quad (4)$$

Thus squared deviation equals expected excess Brier loss relative to that ideal benchmark, without needing to observe the particular realization. Market prices provide an empirically motivated proxy for this conditional probability. The identity states the ideal-reference assumption explicitly; marginal calibration alone does not establish it for each contract. The current analysis measures the proxy directly, while subsequent settlement permits a complementary outcome-based evaluation.

3 Results

3.1 Aggregate and contract-level agreement

Mean absolute deviation is 33.17 percentage points for question-only forecasts and 8.58 points for research forecasts, a paired mean reduction of 24.58 points. Research yields closer agreement in five contracts and identical agreement in the remaining contract (Figure 1). No contract was excluded after forecasting. All twelve responses passed validation on their first attempt; none was repaired or rerun.

The constant-50% reference has a mean absolute deviation of 27.75 points, falling between the two model conditions. Research also has lower root mean squared deviation (11.40 versus 40.83 points) and lower mean distance outside the opening spread (7.67 versus 32.25 points). Table 2 reports all three measures on the same six-contract cohort. Follow-up quotes collected at reveal have unchanged midpoints for every contract, so the midpoint-based results are identical using those later observations. Netflix’s spread widens from three to seven cents.

Table 1. Contract definitions and probabilities.

YES event	Opening market (%)		Forecast (%)	
	Bid–ask	Midpoint	Question only	Research
LAX high 74–75°F (Sep. 18)	3–4	3.5	12	5
US unemployment > 4.4% (Sep.)	1–2	1.5	50	3
Fed holds rates (Oct. 28)	48–49	48.5	55	55
Neutron launches before 2027	4–8	6	65	20
Netflix film ranks first ^a	61–64	62.5	0.5	40
Milwaukee beats Baltimore (Sep. 20)	63–64	63.5	49	58

Notes: All dates are in 2026. ^a*Why Did I Get Married Again?* ranks first in the US Netflix film chart published September 22 for the week ending September 20. The contract headline refers to September 21; settlement rules define the event.

Table 2. Deviation from the opening market benchmark.

Measure (percentage points)	Question only	Research	Constant 50%
Mean absolute gap	33.17	8.58	27.75
Root mean squared gap	40.83	11.40	33.64
Mean distance outside spread	32.25	7.67	26.83

Notes: Equal-weight averages over all six contracts, using the opening midpoint as an intermediate probability benchmark. The constant-50% condition is computed mechanically and requires no additional model calls.



Figure 1. Forecast probabilities and opening Kalshi midpoints by question. Each panel connects the question-only forecast on the left to the outside-research forecast on the right. A labeled horizontal dashed line marks the opening Kalshi midpoint. The vertical distance from each forecast to this line is its absolute market deviation. Boxed slope labels report the reduction in that deviation in percentage points and, in parentheses, as a percentage of the question-only deviation: $100(d_Q - d_R)/d_Q$, where $d_k = |p_k - q|$. Positive values indicate improvement. Endpoint labels report probabilities; all panels share a 0–100% scale. The two forecasts are separate information conditions, not chronological updates.

3.2 Evidence use and observed failure modes

The largest reductions in absolute deviation occur for unemployment (47 points), Neutron (45 points), and Netflix (39.5 points). For unemployment, the BLS reading of 4.1% provides a current anchor: the contract’s strict threshold requires a published rate of at least 4.5%. The forecast falls from 50% to 3%. For Neutron, the company update concerns Q4 delivery to the launch pad, rather than a guaranteed launch that quarter. The forecast falls from 65% to 20%, but remains fourteen points above the market midpoint.

The Netflix case illustrates a limitation of both source selection and model interpretation. Both responses reason as though the target were an older catalog film. Official discovery identified a September 2026 release, but the packet preserved the previous week’s number-one ranking without the release date. This omission and the entity-identification error were recorded at 11:25:43 UTC, before target reveal [7]. The original forecasts remain in the analysis: research raises the estimate from 0.5% to 40%, compared with a midpoint of 62.5%.

Smaller reductions occur for Milwaukee baseball (nine points) and LAX weather (seven points). Team records and run differentials favor Milwaukee, but the packet lacks starting pitchers, lineups, and bullpen context. For LAX, an NWS forecast high of 79°F lies outside the 74–75°F event interval, reducing the forecast from 12% to 5%. The probability assigned to that interval is not supported by an estimated forecast-error distribution. Finally, both Fed forecasts are 55%, despite the researched explanation incorporating the latest policy decision and year-end projections. Equal probabilities therefore do not imply equal factual grounding.

4 Discussion

The results show a substantial improvement against the market benchmark: research reduces the mean absolute gap by 24.58 percentage points and improves agreement in five of six cases. Given the forecasting and calibration evidence for prediction markets, this is a useful intermediate performance result. The design produces feedback while outcomes remain in the future, permitting research methods to be compared without waiting for settlement or introducing realized-outcome leakage through retrospective data collection.

The relevant experimental contrast is access to current evidence. The question-only condition lacks fresh facts that are directly relevant to several events, so a subsequent comparison can distinguish a minimal factual update from deeper, adaptive research. Review, scenario modeling, and ensembles remain additional dimensions to test. Source collection here was performed by an external coding agent; the results concern that research pipeline coupled to the forecasting model.

Several additional limitations matter. First, six purposively selected contracts do not support a population-level performance estimate. Employment and monetary policy are related, and contract horizons differ substantially. Second, one draw per condition leaves model sampling variability unmeasured; the observed paired differences combine information-condition differences with stochastic generation. We report no significance test or inferential interval. Third, the strongest case for a price benchmark concerns liquid, well-functioning markets. This pilot’s minimal two-sided-quote requirement does not establish deep liquidity: Neutron’s opening best bid has a size of only five contracts. Larger studies should prespecify stronger depth and spread criteria and examine performance by horizon, reflecting the empirical calibration evidence [2]. Fourth, research follows the opening snapshots. Unchanged follow-up midpoints reduce concern about observed price movement during the run but do not establish identical information sets. The remaining blinding

limitations concern price exposure, as described in Section 2.3.

The pilot also identifies a concrete design improvement: both information conditions in a subsequent research-depth comparison should include verified entity identities and relevant release or publication dates. A fresh, unrevealed cohort could then compare a small current-facts packet with additional targeted research, using repeated forecasts and registered handling of failed submissions. Larger cohorts would permit paired uncertainty estimates that account for related event groups. Retaining these forecasts through settlement would add outcome scores and calibration checks to the immediate market-based evaluation. These two forms of assessment are complementary: prices support timely development and comparison, while realizations allow the intermediate benchmark and the model forecasts to be evaluated over repeated events.

References

- [1] Wolfers, J., and Zitzewitz, E. (2004). Prediction markets. *Journal of Economic Perspectives*, 18(2), 107–126. doi:10.1257/0895330041371321.
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- [5] Study protocol and experimental registration (2026). Vorhersage blinded Kalshi pilot, September 18. protocol.json, run/selection.json, and run/registration.json.
- [6] Forecasts and market comparison (2026). Vorhersage blinded Kalshi pilot, September 18. run/comparison.json, run/raw-records.json, and run/revealed-targets/.
- [7] Integrity audit and pre-reveal diagnostics (2026). Vorhersage blinded Kalshi pilot, September 18. run/audit.json and run/pre-reveal-diagnostics.json.

A Package architecture and execution

Vorhersage 0.3.0 is a Python forecasting workbench with a command-line interface, requiring Python 3.11 or later and no declared runtime dependencies. It preserves questions, evidence, model inputs, forecasts, and evaluation records. The external agent chooses sources and makes judgments; deterministic package code checks schemas, bookkeeping, and declared numerical calculations. Stored provenance makes a claim inspectable but does not establish that it is true.

A.1 Components and responsibilities

Component	Role in this study
<code>store.py</code> , <code>schemas.py</code>	SQLite transactions, versioned questions, immutable artifact records, content hashes, and validated submissions preserve what each trial received and produced.
<code>experiments.py</code>	Registers a versioned method and separate arms binding method, model configuration, and question-specific evidence. Freezes the cohort, cutoffs, repetitions, and trial order.
<code>evidence.py</code>	Stores source captures, URLs, timestamps, excerpts, facts, and limitations in packets linked to the assigned arm.
<code>workflow.py</code>	Advances each trial through assessment and issuance; validates the submitted probability and rationale, then records the forecast.
<code>market_data.py</code>	Separates price-free contract definitions from private quote snapshots; derives the YES ask from the complementary NO bid.
<code>workbench.py</code>	Keeps targets in a separate evaluator project, stores commitments, and requires completion before reveal.
Study scripts	<code>study.py</code> coordinates selection, source filtering, registration, import, forecast sealing, reveal, and audit. <code>build_evidence.py</code> records the source-to-packet extraction.

A.2 Arm specification and model configuration

The shared method, `outside-source-direct-judgment`, uses only `assessment` and `issue`. Assessment asks for a probability, concise rationale, reasons it could be too high or too low, and limitations. Issuance records the result; it is not another model call. The question-only arm assigns no research packet, while the outside-research arm assigns its question’s frozen packet. Changing a registered method or arm requires a new version.

Both arms used temperature 0.5, topK 40, topP 1, `maxOutputTokens` 8192, and `thinking_budget` 2048. The model was called through an external EDSL transport with tools disabled. This transport and the report’s plotting dependencies are separate from the dependency-free core. Prompts, provider records, raw answers, accepted submissions, and issued forecasts were all retained. Model identity and settings are those recorded by the transport, not an attestation about the provider’s internal implementation.

A.3 Scope of the evaluated implementation

The package also supports scenario mixtures, grouped likelihood-ratio updates, timeline models, reference-class priors, review stages, ensembles, and scoring against resolved outcomes. Those

features were not used to produce these twelve forecasts. Exa and Firecrawl adapters can preserve research-provider receipts, but this run used browser tools and direct official HTTP requests because their API keys were not configured. Thus the result concerns a frozen-evidence, direct-judgment comparison, not the performance of every available workflow.

A.4 Integrity checks and recorded costs

The live audit verified all twelve issued forecasts and six opening-target commitments, with 82 artifacts passing database integrity checks. Fourteen focused tests covered source restrictions, quoted-odds rejection, premature or tampered reveal prevention, and metric arithmetic. All twelve model responses were accepted on their first attempt. Summed provider-reported model cost was \$0.16835. Research, browser, and coordinator dollar costs are unknown, not zero; this number is not total system cost. The ledger records fourteen direct HTTP attempts (seven successful), six browser searches, and six browser page-read operations across five tool calls.

B Evidence and reproducibility

B.1 The six frozen evidence packets

The packets contain nine source records. The following links identify the original sources; the dated captures in the run are authoritative for what the model received, since live pages may subsequently change.

Weather. [NWS point forecast](#) and the [LAX climate report](#): forecast high 79°F, previous high 78°F, and normal high 76°F. The packet does not estimate forecast errors and uses NWS as a proxy for the contract's commercial settlement source.

Employment. [BLS employment report, September 4](#): August and July unemployment 4.1%, June 4.2%, and August payroll growth 162,000. The September report was scheduled for October 2. The contract's strict threshold requires a published rate of at least 4.5%.

Monetary policy. [September 16 FOMC statement](#) and [economic projections](#): a 25-basis-point hike to 3.75–4.00%, unanimous 12–0 vote, and a median year-end rate projection of 4.1%. Year-end projections do not identify the meeting at which a subsequent change occurs.

Space technology. [Rocket Lab second-quarter update, August 10](#): first-flight assembly and testing, with first-stage tank delivery to the pad aligned to Q4. The issuer's forward-looking update lacks a complete remaining test and launch-readiness schedule.

Entertainment. [Netflix US film chart](#): the target title ranked first for September 7–13. Its [September 9 release date](#) was found during discovery but omitted from the model's packet. The error was logged in `run/pre-reveal-diagnostics.json` at 11:25:43 UTC.

Sports. [MLB schedule](#) and [standings](#): Milwaukee at Baltimore on September 20; records of 95–58 and 75–78, with run differentials of +197 and –26. No usable starting-pitcher, lineup, injury, or bullpen context entered the packet.

B.2 Research artifacts and reproducibility

Paths are relative to `examples/kalshi_blind_20260918/`. The protocol and candidate order preceded price inspection; experiment registration followed evidence collection at 11:19:02 UTC.

Record	What it establishes
<code>protocol.json</code> ; <code>run/selection.json</code>	Planned comparison, source rules, and ordered candidates.
<code>run/registration.json</code> ; <code>run/*-arm.json</code>	Frozen questions, method/model/data assignments, and trials.
<code>run/packets/</code> ; <code>run/tasks/</code>	Exact evidence and prompts available to each call.
<code>run/raw-records.json</code> ; <code>run/attempts.json</code>	Responses, attempt outcomes, and model costs.
<code>run/sealed-forecasts.json</code> ; <code>run/audit.json</code>	Forecast seal, hash checks, and timing.
<code>run/comparison.json</code> ; <code>run/revealed-targets/</code>	Numerical results and original revealed target receipts.

`plot_performance.py` recomputes all three aggregate metrics for all three conditions and verifies them against the frozen comparison before generating the table rows and calling `plot_performance.R` for the faceted slope graph. The R script also verifies the metrics and renders the figure using `ggplot2`. With Python, R (`ggplot2` 3.5.0 or later and `jsonlite`), and LaTeX available, run from this study directory:

```
python plot_performance.py
latexmk -pdf -interaction=nonstopmode -halt-on-error report.tex
```

Rebuilding makes no model or market calls. A new blind experiment requires unrevealed contracts and a separate directory preserving the registration, sealing, and reveal order.